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A discrete memristive cyclic Hopfield neural network with multi-cavity-like attractors and application in secure communication

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ABSTRACT

Discrete memristors with synapse-like properties play a significant role in elucidating the complex neurodynamic mechanisms of biological neural networks in the brain. This work presents a discrete memristive cyclic Hopfield neural network (DMCHNN), which integrates a novel discrete cosine memristor model for simulating mutual synaptic connections with a cyclic Hopfield neural network. Numerical analysis indicates that the system can control the chaotic range through amplitude modulation of the coupling strength k and weight w_{31} , generating different numbers of multi-cavity-like hyperchaotic attractors under the regulation of k . By varying the initial value of the memristor, the system exhibits homogeneous and heterogeneous initial offset-boosting behaviors, accompanied by the coexistence of numerous homogeneous and heterogeneous attractors. Entropy performance evaluation further confirms that the system possesses excellent randomness. Furthermore, multi-cavity-like attractors produced by the system are successfully implemented on an FPGA hardware platform. Experimental results of the HE-DCSK communication system constructed based on DMCHNN demonstrate superior anti-noise performance compared to other chaotic maps.

1. Introduction

In the study of biological neural networks, revealing neurodynamic behavior is crucial for gaining an in-depth understanding of the brain's internal information processing mechanisms (Vignesh et al., 2025). As a representative class of artificial neural network models, the Hopfield neural network (HNN), with its unique topological structure, nonlinear activation functions, and rich dynamic characteristics, demonstrates important theoretical value in simulating brain functions such as associative memory and dynamic pattern formation (Hopfield, 1984; Yang et al., 2017; Yu et al., 2024). This network has not only become a fundamental model in the study of nonlinear dynamics and neural networks but also garnered widespread attention due to its ability to effectively exhibit complex dynamic characteristics such as chaotic behavior (Li et al., 2025) and synchronization phenomena (Min et al., 2025). In recent years, to accommodate different application scenarios and theoretical needs, researchers have proposed various improved variants of the HNN, such as the discrete heterogeneous HNN (Yang et al., 2025b), the fractional-order heterogeneous HNN (Luo et al., 2025b), and the fractional-order HNN with time delays (Nie et al., 2021), further expanding its research depth in neurodynamics analysis and system modeling.

Memristors, as electronic components characterized by memory, nonlinearity, and synaptic-like properties, are widely used in fields such as bio-inspired circuits (Deng et al., 2025a; Shi et al., 2025), neuromorphic computing (Duan et al., 2024), chaotic systems (Deng et al., 2025b; Wang et al., 2025c; Yang et al., 2025a), and other fields (Liu et al., 2026; Peng et al., 2025b; Qin & Lai, 2026). The discrete memristor is a model derived from the discretization of the continuous memristor, fully retains the essential characteristics of the memristor. Compared to continuous models, discrete memristors offer notable advantages such as reduced computational time and greater ease of implementation in digital hardware (Deng et al., 2025c; Xu et al., 2024). Furthermore, introducing discrete memristors into classical chaotic systems can effectively enhance the chaotic dynamic behaviors of these systems—for instance, by expanding chaotic regions and increasing sequence complexity (Wang et al., 2024; Zhang et al., 2025a). These properties underscore the significant research value and application potential of discrete memristors in theoretical exploration and engineering implementation. Luo et al. (2025a) constructed three hyperchaotic maps with ultra-high Lyapunov exponents by introducing nonlinear oscillation terms into three discrete memristors. Inspired by the integration of time delay and discrete memristors, Liu et al. (2025) proposed a general discrete time-delayed memristor model and developed four discrete time-delayed

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hyperchaotic maps with hidden attractors. Hussan et al. (2025) designed a dual-memristor hyperchaotic map capable of generating numerous hyperchaotic attractors by connecting two discrete memristors in a cascaded feedback approach. Ma et al. (2025) utilized a discrete locally active memristor to drive a neural map, inducing heterogeneous coexistence and multistable behavior.

The HNN has become an important model in nonlinear dynamics research due to its rich chaotic dynamic characteristics. Researchers have extensively explored the dynamic behaviors of HNNs with different topological structures (He et al., 2025; Yu et al., 2025a). For instance, to overcome the dynamic limitations of simple unidirectional topologies, Wang et al. (2025a) constructed a tri-memristor hyperchaotic ring neural network with hidden chaos and homogeneous/heterogeneous multistability by introducing three memristors into a ring HNN. Hu et al. (2024) integrated two memristive HNNs into a single memristive Hopfield switching neural network by employing a weight-switching mechanism. The designed switching function enables the generation of multi-scroll chaotic attractors. Lai and Guo (2024) developed a central cyclic memristive neural network by connecting three neurons to a single Hopfield neuron as the central node and introducing a memristor as the self-connection synapse. A large-range amplitude control and homogeneous/heterogeneous coexisting attractors were observed in the system. Additionally, the team constructed a simplified cyclic HNN capable of exhibiting multiple firing patterns (Lai et al., 2025), based on a unidirectional cyclic HNN without self-synapses, incorporating external current stimulation and electronic radiation simulated by memristors. In recent years, research on the neurodynamics of HNNs in the discrete domain has gradually gained attention. Bao et al. (2024) proposed a simplified two-dimensional discrete HNN model without self-connections by discretizing a continuous HNN, which exhibits coexisting multiple attractors and polyhedral hyperchaotic attractors. Wei et al. (2025) constructed a discrete memristive HNN capable of generating multi-ringlike attractors by embedding a discrete memristor as a self-synapse into a two-neuron HNN. Bao et al. (2025) further designed a discrete HNN with the memristor initial-boosting mechanism by replacing the original fixed resistor with an adaptive discrete memristor weight. Yu et al. (2025b) developed a discrete memristive HNN with electromagnetic coupling by introducing two memristors as an electronic radiation source and a synapse coupling. The system can produce an attractor growth phenomenon. However, the neurodynamics study of discrete memristive HNNs with specific topological connection patterns remains relatively scarce and has not yet been reported in the literature.

Inspired by the aforementioned studies, this paper introduces a discrete memristor as a synapse into a three-neuron unidirectional cyclic HNN, constructing a DMCHNN. Through dynamic analysis methods such as Lyapunov exponent (LE) spectra, bifurcation diagrams, and phase portraits, complex dynamic behaviors of the system are unveiled, including hyperchaos, amplitude control, multi-cavity-like attractors, and homogeneous/heterogeneous initial offset-boosting. Besides, the circuit design and hardware implementation of the system are completed based on an FPGA platform. In terms of application, a DMCHNN-based HE-DCSK system scheme is proposed, and its anti-noise performance is tested.

The main contributions of this paper are as follows:

- 1) A novel DMCHNN is presented by introducing a discrete memristor into a cyclic HNN. Theoretical analysis demonstrates that the fixed point set of the system contains innumerable unstable points.
- 2) Numerical results reveal that the system exhibits complex dynamic characteristics, including hyperchaos, amplitude control, multi-cavity-like hyperchaotic attractors, and homogeneous/heterogeneous initial offset-boosting behaviors.
- 3) Based on the FPGA hardware platform, the DMCHNN is successfully implemented and applied to a HE-DCSK system. Experimental results show that this scheme exhibits superior anti-noise performance compared to other chaotic map-based communication systems.

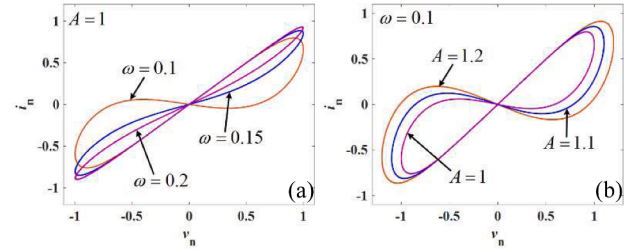


Fig. 1. Pinched hysteresis loops of the discrete memristor. (a) $\omega = 0.2, 0.15$, and 0.1 for $A = 1$. (b) $A = 1.2, 1.1$, and 1 for $\omega = 0.1$.

The subsequent structure of this paper is organized as follows: Section 2 elaborates on the mathematical model of DMCHNN and analyzes its fixed points and stability. Section 3 investigates the dynamic behavior of the DMCHNN in detail. Section 4 designs and implements the FPGA hardware circuit for DMCHNN. Meanwhile, a HE-DCSK system based on DMCHNN is constructed, and its performance is evaluated. Finally, Section 5 summarizes the entire paper and outlook on future research directions.

2. Mathematical modeling of DMCHNN

2.1. Discrete memristor model

Based on the fundamental definition of the memristor, the discrete memristor is obtained by discretizing the continuous memristor using the forward Euler method. Accordingly, we propose a discrete cosine memristor model, whose mathematical expression is as follows:

$$\begin{cases} i_n = W(\varphi_n)v_n = \cos(\varphi_n)v_n \\ \varphi_{n+1} = \varphi_n + \varepsilon \sin(v_n) \end{cases} \quad (1)$$

where i_n , v_n , and φ_n represent the discrete current, voltage, and flux at the n th time iteration, respectively. ε denotes the parameter of the memristor. $W(\varphi_n) = \cos(\varphi_n)$ is the discrete memductance.

By applying a discrete voltage source $v_n = A \sin(\omega n)$ to the discrete memristor with parameter $\varepsilon = 0.1$, the hysteretic characteristics of the memristor can be observed. When the amplitude $A = 1$, the pinched hysteresis loops corresponding to frequencies $\omega = 0.2, 0.15$, and 0.1 on the $v_n - i_n$ plane are shown in Fig. 1(a). With the frequency fixed at $\omega = 0.1$, Fig. 1(b) displays the pinched hysteresis loops for amplitudes $A = 1.2, 1.1$, and 1 . As illustrated in Fig. 1, it can be observed that as the frequency ω increases, the lobe area of the hysteresis loop decreases; conversely, as the amplitude A increases, the lobe area expands. This behavior aligns with the fingerprint features and basic definition of the memristor.

2.2. DMCHNN model

The HNN is a single-layer, fully connected, and recurrent artificial neural network. It exhibits rich intrinsic dynamic characteristics and can effectively simulate the firing activities of biological neurons as well as certain information processing mechanisms in the brain. Due to these properties, researchers have extensively investigated HNNs with various connection structures and their dynamic behaviors, including ring-structured and cyclic connection topologies. Based on the characteristics of the cyclic HNN and the discrete modeling method, we introduce the proposed discrete memristor as a synaptic element into a three-neuron cyclic HNN, thereby constructing three distinct DMCHNNs. Fig. 2 presents three different topological connections of DMCHNN, and their mathematical formulation are described as follows:

$$\begin{cases} x(n+1) = \alpha x(n) + k \cos(u(n)) \sin(y(n)) \\ y(n+1) = \beta y(n) + w_{23} \sin(z(n)) \\ z(n+1) = \gamma z(n) + w_{31} \sin(x(n)) \\ u(n+1) = u(n) + \varepsilon \sin(\sin(y(n))) \end{cases} \quad (2)$$

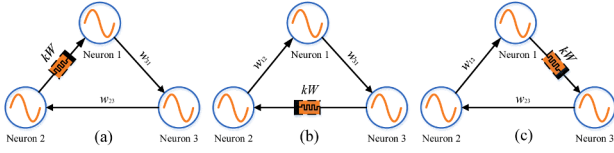


Fig. 2. Three different topological connections of DMCHNN. (a) The memristor is coupled between neurons 1 and 2. (b) The memristor is coupled between neurons 2 and 3. (c) The memristor is coupled between neurons 1 and 3.

$$\begin{cases} x(n+1) = \alpha x(n) + w_{12} \sin(y(n)) \\ y(n+1) = \beta y(n) + k \cos(u(n)) \sin(z(n)) \\ z(n+1) = \gamma z(n) + w_{31} \sin(x(n)) \\ u(n+1) = u(n) + \varepsilon \sin(\sin(z(n))) \end{cases} \quad (3)$$

$$\begin{cases} x(n+1) = \alpha x(n) + w_{12} \sin(y(n)) \\ y(n+1) = \beta y(n) + w_{23} \sin(z(n)) \\ z(n+1) = \gamma z(n) + k \cos(u(n)) \sin(x(n)) \\ u(n+1) = u(n) + \varepsilon \sin(\sin(x(n))) \end{cases} \quad (4)$$

where α , β , and γ are all the neural decay coefficients. k represents the coupling strength. w_{12} , w_{23} , and w_{31} denote the connection weights. ε is the internal parameter of the discrete memristor. $x(n)$, $y(n)$, and $z(n)$ represent the membrane potentials of the neurons at the n th time iteration.

As illustrated in Fig. 2, the discrete memristor is coupled between different neurons, forming DMCHNNs with three different topological structures. Due to differences in their topological structures, these three systems exhibit distinct differences in their dynamical properties, including stability conditions, Lyapunov exponents, attractor morphologies, and the range of chaotic intervals. To delve into the dynamic behaviors of the DMCHNN, this study selects the DMCHNN with memristors coupled between neurons 1 and 2 as the research subject, revealing the existence of its multi-cavity-like attractors.

2.3. Fixed point and stability analysis

The stability of the DMCHNN can be evaluated by analyzing the eigenvalues of its Jacobian matrix at the fixed points. Let P be a fixed point denoted as (X, Y, Z, U) . Its value can be determined by solving the following equation:

$$\begin{cases} X = \alpha X + k \cos(U) \sin(Y) \\ Y = \beta Y + w_{23} \sin(Z) \\ Z = \gamma Z + w_{31} \sin(X) \\ U = U + \varepsilon \sin(\sin(Y)) \end{cases} \quad (5)$$

if all parameters are non-zero, the solution of Eq. (5) is determined by $X = Y = Z = 0$ and U being an arbitrary real number. Consequently, the set of fixed points is $\{(0, 0, 0, m) | m \in \mathcal{R}\}$, indicating that the DMCHNN possesses infinitely many fixed points.

With the parameters set as $(\alpha, \beta, \gamma, k, w_{23}, w_{31}, \varepsilon) = (0.6, 0.7, 0.5, 5, 10, 1, 0.1)$, the Jacobian matrix at the fixed point P can be expressed as:

$$J = \begin{bmatrix} 0.6 & 5F_2(U)F_2(Y) & 0 & -5F_1(U)F_1(Y) \\ 0 & 0.7 & 10F_2(Z) & 0 \\ F_2(X) & 0 & 0.5 & 0 \\ 0 & 0.1F_2(F_1(Y))F_2(Y) & 0 & 1 \end{bmatrix} \quad (6)$$

where the functions $F_1(\cdot)$ and $F_2(\cdot)$ are defined as $\sin(\cdot)$ and $\cos(\cdot)$, respectively.

Since the system's fixed point set is $\{(0, 0, 0, m) | m \in \mathcal{R}\}$, these fixed points exhibit a linear distribution pattern in space. Substituting

Table 1

Parameter settings, fixed points, corresponding Jacobian matrix eigenvalues, and stability of the DMCHNN.

Parameter settings	Fixed Points	Eigenvalues	Stability
$\alpha = 0.6,$	$(0, 0, 0, 0)$	$(4.2849, -1.2425 \pm 3.1897i, 1)$	Unstable
$\beta = 0.7,$	$(0, 0, 0, \pm 0.5)$	$(4.1281, -1.1640 \pm 3.0538i, 1)$	Unstable
$\gamma = 0.5,$	$(0, 0, 0, \pm 1)$	$(3.6017, -0.9008 \pm 2.5976i, 1)$	Unstable
$k = 5,$	$(0, 0, 0, \pm 1)$	$(3.6017, -0.9008 \pm 2.5976i, 1)$	Unstable
$w_{23} = 10,$	$\vdots$	$\vdots$	$\vdots$
$w_{31} = 1,$	$\vdots$	$\vdots$	$\vdots$
$\varepsilon = 0.1$	$\vdots$	$\vdots$	$\vdots$

$(X, Y, Z, U) = (0, 0, 0, m)$ into Eq. (6), the following characteristic polynomial can be derived:

$$\det(J) = (\lambda - 1)[(\lambda - 0.6)(\lambda - 0.7)(\lambda - 0.5) - 50 \cos m] \quad (7)$$

From this, $|\lambda_1| = 1$ can be obtained. Since $|\lambda_1| = 1$ always remains on the unit circle, if $|\lambda_2| < 1$, $|\lambda_3| < 1$, and $|\lambda_4| < 1$, then the fixed point is critically stable. If $|\lambda_2| > 1$ or $|\lambda_3| > 1$ or $|\lambda_4| > 1$, then the fixed point is unstable (Bao et al., 2025; Zhang et al., 2023). Table 1 provides a detailed listing of fixed points along with their corresponding Jacobian matrix eigenvalues and stability analysis results. As can be seen from the table, for each fixed point listed, at least one of its corresponding eigenvalues has an absolute modulus greater than 1. Thus, all these listed fixed points are unstable. For other values of m , the fixed points of the DMCHNN may be critically stable or unstable.

3. Dynamic analysis of DMCHNN

To ensure that DMCHNN exhibits rich dynamic behavior within a reasonable parameter range, we fixed the system parameters $(\alpha, \beta, \gamma, w_{23}, \varepsilon)$ to $(0.6, 0.7, 0.5, 10, 0.1)$ through extensive numerical experiments. The coupling strength k is a key parameter for the connection between the discrete memristor and the HNN, and the weights also significantly influence the dynamic characteristics of the neural network. Thus, we systematically investigated the effects of coupling strength k and weight w_{31} on the system's dynamic behavior. The analysis reveals intricate dynamic phenomena, including the emergence of multi-cavity-like hyperchaotic attractors and homogeneous/heterogeneous initial offset-boosting. Furthermore, the entropy performance of the system is evaluated. Note that we employ a Jacobian matrix method based on QR decomposition to compute the system's LEs, which offers fast computation, numerical stability, and accuracy.

3.1. Dynamics influenced by coupling strength k and weight w_{31}

With the weight fixed at $w_{31} = 1$ and the initial condition set as $(x_0, y_0, z_0, u_0) = (0.1, 0.1, 0.1, 0)$, the corresponding LE spectrum and bifurcation diagram of the DMCHNN are shown in Fig. 3(a) as the coupling strength k varies within the range $[0, 200]$. It can be seen that the system first transitions from a periodic state to a quasi-periodic state. With the emergence of a positive LE, the system enters a chaotic state, and subsequently evolves further into a hyperchaotic state. Within the interval $[0.165, 0.275]$, a decline in the LEs occurs over three time segments due to transient chaotic behavior. Ultimately, the system remains in a sustained hyperchaotic state. Notably, when $k \geq 1.46$, all LEs exhibit an upward trend. Meanwhile, the distribution range of bifurcation points in the x -direction also expands, indicating an enhancement in the system's chaotic behavior. Setting the coupling strength $k = 5$, the influence of weight w_{31} on the system dynamics is investigated, with the results presented in Fig. 3(b). As w_{31} increases, the LEs of the system generally rise, along with a significant expansion in the distribution range of bifurcation points in the z -direction, demonstrating a further increase in the dynamical complexity of the system. Meanwhile, the results shown in Fig. 3 indicate that the system can sustain a hyperchaotic state over a wide range of variations in parameters k and w_{31} , and the overall LEs

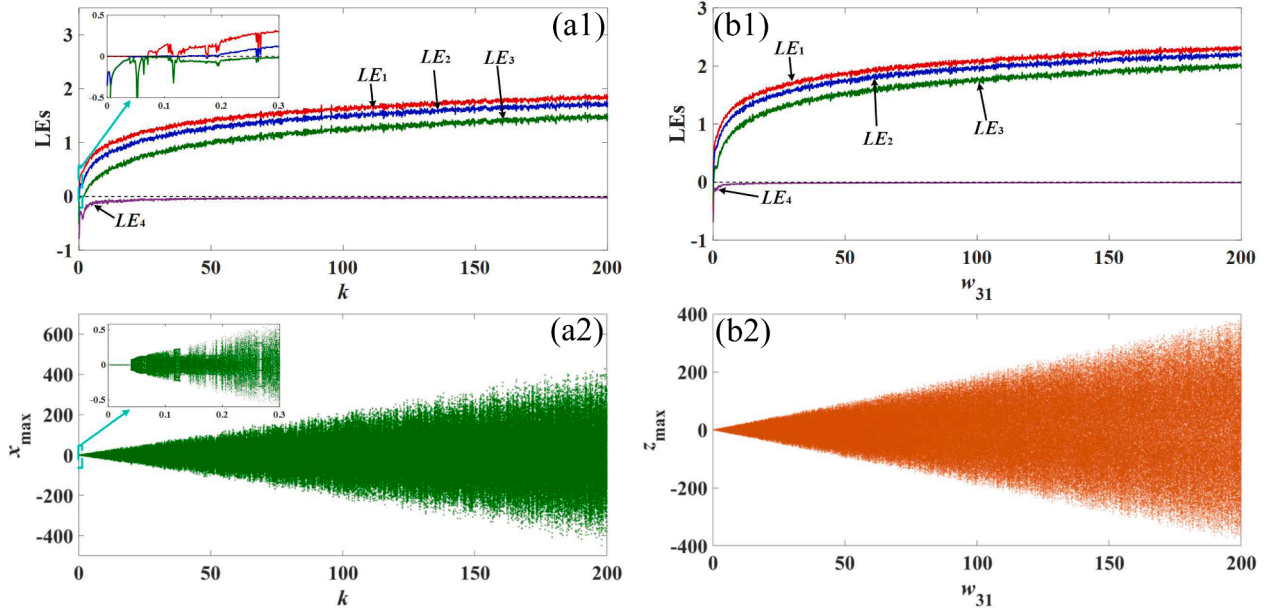


Fig. 3. Dynamical behaviors influenced by the coupling strength k and the weight w_{31} . (a) LE spectrum and bifurcation diagram with $k \in [0, 200]$ and $w_{31} = 1$. (b) LE spectrum and bifurcation diagram with $w_{31} \in [0, 200]$ and $k = 5$.

exhibit an upward trend. From an application perspective, this broad chaotic parameter range provides a larger parameter selection space for chaotic communication or encryption systems, which helps enhance the adaptability and security of the system (Wang et al., 2025b).

Under the influence of the coupling strength k , the DMCHNN generates multiple firing patterns, including periodic spiking, chaotic spiking, and chaotic bursting, as shown in Fig. 4. Fig. 4(a) and (b) display two different periodic spiking patterns. As the coupling strength k increases, the periodic spiking pattern transitions into a chaotic spiking pattern. When k is further increased, another chaotic bursting pattern emerges.

Moreover, the bifurcation diagrams illustrated in Fig. 3 indicate that the oscillation amplitude of the system is effectively modulated by the coupling strength k and the weight w_{31} . To validate this amplitude control characteristic, we selected values of $k = 5, 25, 45, 65,$ and 85 , and plotted the corresponding hyperchaotic attractors in the $x - y$ plane, as shown in Fig. 5(a). It can be observed that as k increases, the spatial extent of the attractor expands significantly along the x -direction. Similarly, Fig. 5(b) displays the phase portraits in the $y - z$ plane for $w_{31} = 10, 30, 50, 70,$ and 90 . It can be observed that the amplitude of the attractor in the z -direction also increases. This ability to achieve amplitude control through parameter modulation not only effectively regulates the system's chaotic range but also highlights the DMCHNN's reliable stability and rich dynamical complexity.

3.2. Multi-cavity-like hyperchaotic attractors

Under the condition that the weight w_{31} and the initial values remain consistent with those in Fig. 3(a), and with the number of iterations set to 20000, a series of multi-cavity-like attractors can be obtained on the $u - x$ plane by varying the coupling strength k , as plotted in Fig. 6. It is noteworthy that as the number of iterations further increases, the number of multi-cavity-like attractors in Fig. 6 also rises correspondingly. These attractors are termed multi-cavity-like due to their topological resemblance to cavities, where the cavities are closely interconnected, and the phase trajectories randomly jump between different cavities, as shown in Fig. 7. Unlike traditional multi-cavity attractors (Li et al., 2024), the individual cavities here exhibit variations in size, particularly in the amplitude range along the x -direction. This characteristic aligns with the amplitude control effect described in Section 3.1, further confirming the tenability of the coupling strength k on the system's

dynamic behavior. Additionally, the number of cavities varies with the coupling strength k . For instance, when $k = 7.3$, the system only produces a single-cavity-like attractor, whereas a double-cavity-like attractor emerges under $k = 2.9$. This phenomenon uncovers the complexity of the system's dynamics as the coupling strength k evolves. Combined with the LE spectrum in Fig. 3(a), it is evident that all these multi-cavity-like attractors are in a hyperchaotic state.

As shown in Fig. 6, under the control of parameter k , the number of cavities in the multi-cavity-like attractors generated by the system exhibits non-deterministic variation characteristics. This phenomenon primarily stems from the use of sine and cosine functions for both the memristor and activation function in the system. As the parameter k increases, due to the boundedness of trigonometric functions, the evolution range of the system state variables in the x -direction expands, causing the cavity structure's size in the x -direction to gradually increase. Meanwhile, changes in the number of cavities are influenced by the complex interactions among state variables during the system's nonlinear iterative process. Consequently, the variation in the number of multi-cavity-like attractors generated by DMCHNN is non-deterministic. The diverse and complex multi-cavity-like hyperchaotic attractors generated by DMCHNN may reflect more intricate dynamic behaviors or electrical activity states in simulated biological neural networks (Zhang et al., 2025b, 2021).

3.3. Homogeneous/heterogeneous initial offset-boosting

To investigate the influence of initial conditions on the dynamical behavior of the DMCHNN, we set the coupling strength k and the weight w_{31} to 0.1 and 1, respectively, and choose the initial values $(x_0, y_0, z_0, u_0) = (0.1, 0.1, 0.1, u_0)$, where the initial value u_0 of the memristor serves as a variable parameter. Fig. 8 illustrates the memristor initial-dependent LE spectrum and bifurcation diagram. As u_0 varies within the interval $[0, 8\pi]$, the system's LEs exhibit three distinct states: hyperchaos, chaos, and periodicity. The dense bifurcation regions and the fixed point distribution regions in the corresponding bifurcation diagram are consistent with the LE spectrum. Moreover, with u_0 increasing, the bifurcation blocks shift entirely along the u -direction, demonstrating a pronounced initial offset-boosting behavior.

To further confirm this phenomenon, we select three sets of initial conditions from values of u_0 , corresponding to hyperchaotic, chaotic,

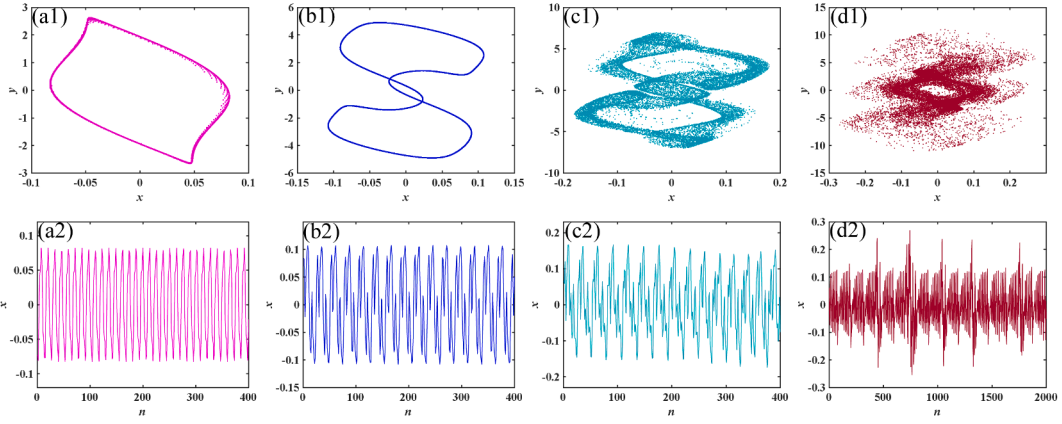


Fig. 4. For $w_{31} = 1$, under the influence of the coupling strength k , the system exhibits diverse discharge patterns. (a) Periodic spiking with $k = 0.045$. (b) Periodic spiking with $k = 0.05$. (c) Chaotic spiking with $k = 0.085$. (d) Chaotic bursting with $k = 0.13$.

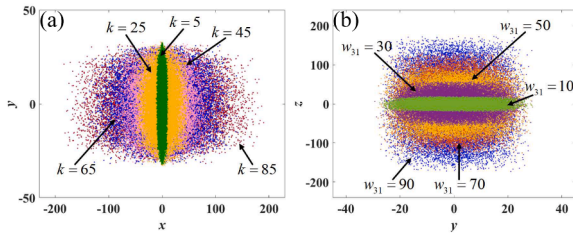


Fig. 5. The amplitude of the system's attractors is modulated by the coupling strength k and the weight w_{31} . (a) For $w_{31} = 1$, attractors in the $x - y$ plane for different values of k : 5 (green), 25 (yellow), 45 (purple), 65 (blue), and 85 (brown). (b) For $k = 5$, attractors in the $y - z$ plane for different values of w_{31} : 10 (green), 30 (purple), 50 (yellow), 70 (orange), and 90 (blue).

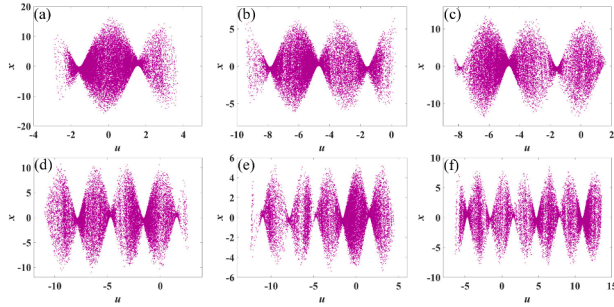


Fig. 6. For $w_{31} = 1$, multi-cavity-like attractors under different coupling strength k . (a) 1-cavity-like attractor with $k = 7.3$. (b) 2-cavity-like attractor with $k = 2.9$. (c) 3-cavity-like attractor with $k = 6.1$. (d) 4-cavity-like attractor with $k = 5$. (e) 5-cavity-like attractor with $k = 2.4$. (f) 6-cavity-like attractor with $k = 3.7$.

and periodic states, respectively. Their attractors are shown in Fig. 9. It can be observed that three heterogeneous attractors with different structures coexist on the $u - x$ plane. When u_0 increases by $2n\pi$ ($n = 1, 2, 3$), these heterogeneous attractors synchronously shift along the u -direction, generating homogeneous attractors with identical structures. Numerical results indicate that by adjusting the initial value u_0 of the memristor, the DMCHNN can achieve coexistence of homogeneous and heterogeneous attractors. This behavior reflects the system's high sensitivity to initial conditions and its rich multistable dynamics. Specifically, the heterogeneous initial offset-boosting behavior further enhances the neural network's complexity, robustness, and stability. Meanwhile, it endows the system with greater adaptability, granting it strong developmental potential in complex and dynamic application scenarios (Lai & Guo, 2024).

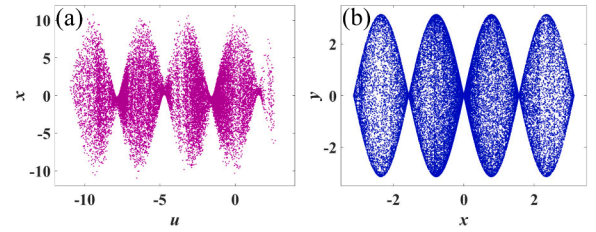


Fig. 7. Comparison of the multi-cavity-like and multi-cavity attractors. (a) 4-cavity-like. (b) 4-cavity.

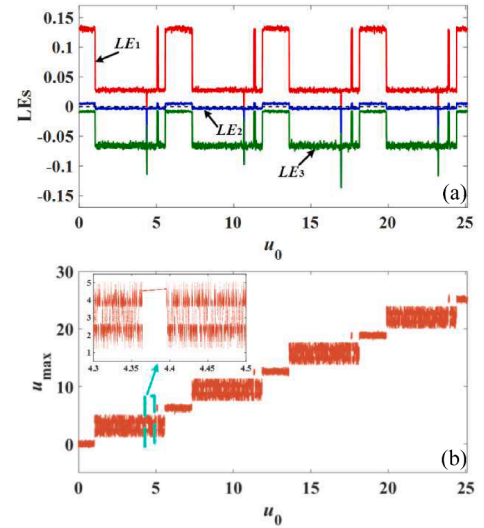


Fig. 8. Memristor initial-related LE spectrum and bifurcation diagram for $k = 0.1$ and $w_{31} = 1$. (a) LE spectrum with $u_0 \in [0, 8\pi]$. (b) Bifurcation diagram with $u_0 \in [0, 8\pi]$.

3.4. Entropy performance evaluation

Permutation entropy (PE), as a nonlinear dynamic metric, can be used to quantify the complexity and regularity of time series. Its core principle lies in analyzing the relative order of adjacent data points within a sequence. Its calculation formula is as follows:

$$PE = \frac{-\sum_{k=1}^{n!} P(x_k) \log_2 P(x_k)}{n!} \quad (8)$$

where n denotes the sequence length. x_k represents the sequence to be tested, and $P(x_k)$ is the probability value of the k th permutation. Gener-

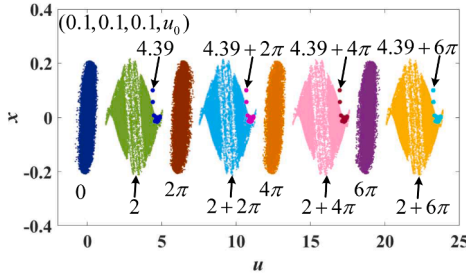


Fig. 9. For $k = 0.1$ and $w_{31} = 1$, homogeneous and heterogeneous initial offset-boosting behavior regarding the initial value u_0 .

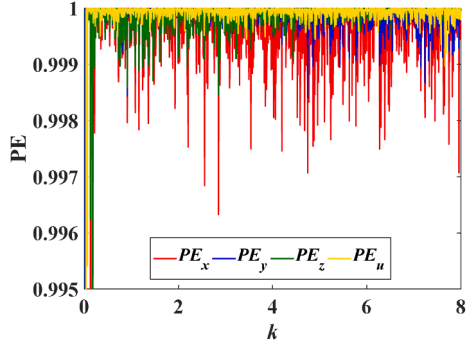


Fig. 10. For $w_{31} = 1$, PE test results for the four chaotic sequences of the DMCHNN with $k \in [0, 8]$.

ally, a higher PE value means that the time series is more complex, random, and irregular. Thus, PE is employed to evaluate the performance of the DMCHNN. Under the condition of initial values set to $(0.1, 0.1, 0.1, 0)$ and $w_{31} = 1$, the variation of PE for the four chaotic sequences when k varies within the range $[0, 8]$ is plotted in Fig. 10. The results indicate that the PE values for each sequence are all greater than 0.996, except when k is in the interval $[0, 0.2]$. The decrease in PE values within this interval can be attributed to the presence of multiple periodic windows in the system, as referenced in Fig. 3(a).

The stochastic performance of a chaotic system serves as a critical foundation for its applications in fields such as secure communication, image encryption (Peng et al., 2025a), and random number generation. Highly random chaotic sequences exhibit stronger unpredictability and superior statistical properties, enabling effective resistance against attacks based on statistical analysis and enhancing system security. To comprehensively evaluate the randomness quality of sequences generated by the DMCHNN, we employ multiple entropy measures for an integrated test, including Approximate entropy (AE), Fuzzy entropy (FE), Increment entropy (IE), Kolmogorov entropy (KE), Sample entropy (SE). These entropy measures quantify the sequence's complexity, irregularity, and randomness from different perspectives. The length of the tested time sequences is 10000. Table 2 lists the parameter settings, initial conditions, and entropy results of the compared chaotic maps. All parameters and initial values are sourced from the original references and have been validated. It can be seen that the DMCHNN outperforms the other chaotic maps across all entropy metrics, particularly with KE and SE being greater than 2. These evaluation results demonstrate that the DMCHNN can generate chaotic sequences with high randomness, making it suitable for engineering applications requiring high levels of randomness, such as secure communication.

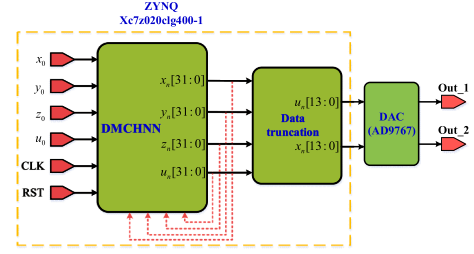


Fig. 11. FPGA-based implementation flow of DMCHNN.

4. Hardware implementation and application in secure communication

4.1. FPGA implementation

Currently, circuit verification of chaotic systems primarily employs two implementation approaches: analog circuits and digital circuits. Given that chaotic maps are inherently discrete, their circuit design can be directly realized using digital circuits. As a typical digital circuit platform, FPGA offers advantages such as high stability, strong flexibility, and prominent parallel computing capabilities. Thus, FPGA is selected for the design and hardware implementation of the DMCHNN.

In the experimental process, the adopted FPGA platform mainly consisted of a Xilinx xc7z020clg400-1 development board, an AD9769 digital-to-analog converter, and an oscilloscope. Fig. 11 presents the implementation flow of the FPGA-based DMCHNN. The DMCHNN module is responsible for performing iterative computations. Its state variables (x_n, y_n, z_n, u_n) are represented in fixed-point format, specifically configured with 1 sign bit, 6 integer bits, and 25 fractional bits. The trigonometric functions are implemented using the CORDIC algorithm (Mohamed et al., 2022). The core formulas of the algorithm are as follows:

$$\begin{cases} O_{i+1} = k_n(O_i - \mu\sigma_i 2^{-i} V_i) \\ V_{i+1} = k_n(V_i + \sigma_i 2^{-i} O_i) \\ W_{i+1} = W_i - \sigma_i \theta_i \end{cases} \quad (9)$$

$$k_n = \prod_{i=0}^{n-1} \sqrt{1 + \mu 2^{-2i}} \quad (10)$$

$$\theta_i = \begin{cases} \operatorname{arctanh}(2^{-i}), & \text{if } \mu = -1 \\ 2^{-i}, & \text{if } \mu = 0 \\ \operatorname{arctan}(2^{-i}), & \text{if } \mu = 1 \end{cases} \quad (11)$$

where k_n and n represent the scaling factor and the number of iterations, respectively. σ_i and θ_i denote the direction and angle of the i th rotation.

The data truncation module is used to adapt the 32-bit data output from the DMCHNN module to the input bit width of the DAC module. Specifically, the AD9769 DAC used in this experiment is a dual-channel 14-bit digital-to-analog converter. Therefore, the higher 14 bits are extracted from the original 32-bit data, while the lower 18 fractional bits are discarded. This truncation strategy not only aligns with the input requirements of the DAC but also ensures the validity of the output data. The DAC module then converts the processed digital signals into analog signals, which are output to an oscilloscope. Finally, the oscilloscope captures and displays the phase diagram results.

In the Vivado programming environment, the system's initial values are fixed at $(0.1, 0.1, 0.1, 0)$, and the weight w_{31} is set to 1. The multi-cavity-like attractors generated by the FPGA-based DMCHNN implementation are shown in Fig. 12. Comparing these with the numerical simulation results presented in Fig. 6(a)–(d) reveals that the two exhibit essentially identical shapes, thereby verifying the feasibility and correctness of the proposed system's hardware implementation. Additionally, the resource utilization of the designed system is summarized in Table 3.

Table 2
Entropy performance evaluation of different chaotic maps.

Chaotic Maps	Parameters	Initial Values	AE	FE	IE	KE	SE
DMCHNN	$(k, w_{31}) = (5, 1)$	$(0.1, 0.1, 0.1, 0)$	2.3159	1.8229	4.5402	2.0160	2.1077
Liu's (Liu et al., 2026)	$(a, b, c, d, e, g, k) = (1.3, 1.4, -0.1, 2, 0.1, -0.5, -0.1)$	$(0.1, 0.1, 1, 1)$	1.9185	1.0971	2.8611	0.6998	1.8027
Zhang's (Zhang et al., 2025a)	$(a, b, c, d, e, f, k_1) = (0.3, 0.89, -1.51, 0.3, 0.98, 0.2, 1.25)$	$(0.11, 0.12, 0.13, 0.14)$	2.1716	0.5850	4.3793	1.9766	2.0349
Liu's (Liu et al., 2025)	$(a, b) = (1.46, 0)$	$(0.4, -0.6, -0.3, -0.3)$	2.0460	0.6544	2.7202	0.4313	1.5671
Wang's (Wang et al., 2024)	$(a, k_1, k_2, k_3, k_4) = (1.4, 0.7, 1.6, 0.6, 0.9)$	$(0.1, 0.1, 0.1, 0.1, 0)$	2.0431	0.7408	3.0328	1.2027	1.9889

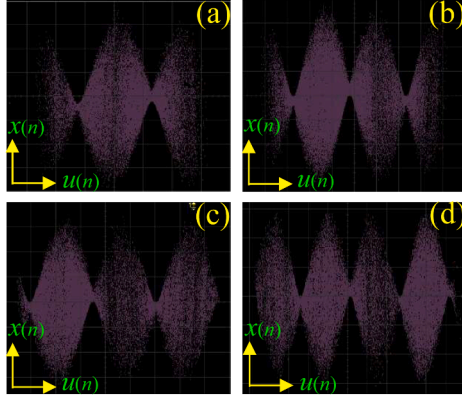


Fig. 12. Multi-cavity-like attractor results generated by the FPGA-based DMCHNN implementation. (a) 1-cavity-like attractor. (b) 2-cavity-like attractor. (c) 3-cavity-like attractor. (d) 4-cavity-like attractor.

Table 3
FPGA resource utilization of the designed system.

Resource	Utilization	Available	Utilization %
LUT	12,129	53,200	7.32
FF	1788	106,400	1.48
DSP	42	220	8.18
IO	34	125	27.20

Table 4
Evaluation results of chaotic sequence generated by DMCHNN in the NIST SP 800-22 test.

No.	Sub-tests	DMCHNN	
		Pass rate	P-value _T
01	Frequency	0.9922	0.8486
02	Block Frequency	0.9766	0.3505
03	Cum. Sums*(F)	0.9922	0.0213
	Cum. Sums*(R)	0.9922	0.8486
04	Runs	1.0000	0.2645
05	Longest Runs	1.0000	0.8195
06	Rank	0.9922	0.6198
07	FFT	0.9688	0.7399
08	Non-Ovla. Temp.*	0.9890	0.4260
09	Ovla. Temp.	0.9766	0.9001
10	Universal	0.9844	0.3115
11	Appr. Entropy	0.9609	0.0372
12	Ran. Exc.*	0.9926	0.5781
13	Ran. Exc. Var.*	0.9821	0.4504
14	Serial(1st)	1.0000	0.8343
	Serial(2nd)	0.9922	0.7565
15	Linear complexity	1.0000	0.3242
	Success No.	15/15	15/15

Note: *The average result of multiple subtests is reported.

The results indicate that the system's utilization rates of LUT, FF, DSP, and IO resources are 7.32%, 1.48%, 8.18%, and 27.20%, respectively, with all indicators falling within acceptable ranges.

Table 5
Evaluation results of chaotic sequence generated by DMCHNN in the TestU01 suite.

Batteries	Length of PRNs	DMCHNN
Rabbit	128 GB	40/40
Alphabit	128 GB	17/17
BlockAlphabit	128 GB	102/102
SmallCrush	6 GB	15/15
Crush	1 TB	144/144

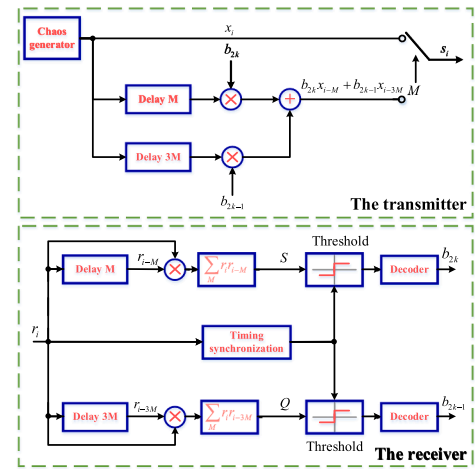


Fig. 13. The overall structure of the transmitter and receiver in the HE-DCSK system.

4.2. DMCHNN-based secure communication

In Section 3, we demonstrate through detailed analysis that the chaotic sequences produced by DMCHNN exhibit outstanding properties in terms of stability, complexity, and randomness, endowing it with significant application potential in secure communication systems. High-Efficiency Differential Chaos Keying (HE-DCSK) serves as a non-coherent chaotic spread-spectrum communication scheme (Yang & Jiang, 2012). Compared to conventional DCSK systems, HE-DCSK achieves twice the bandwidth. Furthermore, due to its non-repetitive signal structure and featureless spectrum, the system exhibits stronger anti-interception capability, thereby enhancing the security of communication data. Building upon these findings, we developed a HE-DCSK system based on DMCHNN and evaluated its noise resistance performance under various channel conditions.

To further evaluate the suitability of the chaotic sequence generated by the DMCHNN for secure communication applications, we conducted two categories of standard randomness tests on it. First, based on the NIST SP 800-22 test suite, we selected 128 binary sequences of length 1,000,000 for evaluation. The system parameters were set to $(k, w_{31}) = (5, 1)$, the initial conditions were $(0.1, 0.1, 0.1, 0)$, and the significance level P -value was set to 0.0001. The results of the 15 tests are presented in Table 4, where all tests were passed successfully, indicating that the sequence exhibits good statistical randomness under NIST standards. Secondly, we performed supplementary validation using the

Table 6

Parameter and initial value settings for different chaotic maps in a secure communication system.

Chaotic Maps	Parameters	Initial Values
DMCHNN	$(k, w_{31}) = (5, 1)$	$(0.1, 0.1, 0.1, 0)$
Liu's (Liu et al., 2026)	$(a, b, c, d, e, g, k) = (1.3, 1.4, -0.1, 2, 0.1, -0.5, -0.1)$	$(0.1, 0.1, 1, 1)$
Xu's (Xu et al., 2024)	$(a, b, c, d, e, k_0, k_1, k_2) = (1.2, 0.1, -1.2, 1.72, \pi/6, 0.1, -10, 0.5)$	$(0.5, 0.5, 0.1, 0.1)$
Qin's (Qin & Lai, 2026)	$(k, k_1, k_2, k_3, k_4, a, b, c) = (3.5, 2.5, 1, -4, 3.4, 0.1, 1)$	$(1, 1, 1, 1, 1)$
Wang's (Wang et al., 2025b)	$(u, w) = (0.3, 1)$	$(0.1, 0.1, 0.1, 0.1, 0.1, 0.1)$
Deng's (Deng et al., 2025b)	$k = 1$	$(0.1, 0.1, 0, 0, 0, 0, 0)$

TestU01 test suite, employing test batteries including Rabbit, Alphabit, BlockAlphabit, SmallCrush, and Crush. According to the TestU01 evaluation criteria, a test is considered passed if its P -value falls within the interval $[0.001, 0.999]$. The test results are summarized in Table 5, indicating that all sub-tests meet the passing criteria. Standardized test results consistently demonstrate that the chaotic sequence generated by DMCHNN exhibits excellent statistical randomness, providing stronger justification for its reliable application in secure communication systems.

The overall structure of the transmitter and receiver in the HE-DCSK system is shown in Fig. 13. At the transmitter, the input signal is generated by a chaos generator. Each frame consists of two time slots: in the k th frame, the first time slot transmits a chaotic reference signal x_i of length M ; the second time slot carries the information bits b_{2k} and b_{2k-1} , which are constructed by multiplying b_{2k} with the chaotic reference signal x_{i-M} from the first time slot of the current frame, and then superimposing the product of b_{2k-1} and the chaotic reference signal x_{i-3M} from the first time slot of the $(k-1)$ th frame. Thus, the transmitted signal s_i for the k th frame can be expressed as follows:

$$s_i = \begin{cases} x_i, & 2kM < i < (2k+1)M \\ b_{2k}x_{i-M} + b_{2k-1}x_{i-3M}, & (2k+1)M < i < 2(k+1)M \end{cases} \quad (12)$$

where M represents the spreading factor.

At the receiver, the received signal r_i consists of two components: the transmitted signal s_i after channel transmission and the superimposed noise signal n_i . To recover the original information bits, r_i is simultaneously fed into two correlators for processing, where correlation operations are performed to achieve signal separation and reconstruction. The detailed information processing flow is described as follows:

$$\begin{aligned} S &= \sum_{i=1}^M r_i r_{i-M} \\ &= \sum_{i=1}^M (s_i + n_i)(s_{i-M} + n_{i-M}) \\ &= \sum_{i=1}^M (b_{2k}x_{i-M} + b_{2k-1}x_{i-3M} + n_i)(x_{i-M} + n_{i-M}) \end{aligned} \quad (13)$$

$$\begin{aligned} Q &= \sum_{i=1}^M r_i r_{i-3M} \\ &= \sum_{i=1}^M (s_i + n_i)(s_{i-3M} + n_{i-3M}) \\ &= \sum_{i=1}^M (b_{2k}x_{i-M} + b_{2k-1}x_{i-3M} + n_i)(x_{i-3M} + n_{i-3M}) \end{aligned} \quad (14)$$

The signals S and Q are sent to the decision devices for detection, and the decision criterion is as follows:

$$\hat{b}_{2k} = \begin{cases} +1, & S > 0 \\ -1, & S \leq 0 \end{cases} \quad (15)$$

$$\hat{b}_{2k-1} = \begin{cases} +1, & Q > 0 \\ -1, & Q \leq 0 \end{cases} \quad (16)$$

In the experiment, we employ DMCHNN and several existing chaotic maps as chaos generators to test their noise resistance performance in the HE-DCSK system. The parameters and initial values for each chaotic map listed in Table 6 adhere to the settings specified in their original

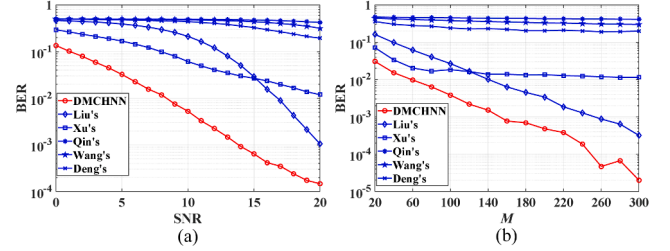


Fig. 14. The BERs of the HE-DCSK system based on different chaotic maps under various (a) SNR $\in \{0, 1, \dots, 20\}$ and $M = 250$. (b) $M \in \{20, 40, \dots, 300\}$ and SNR $= 20$.

literature. The chaotic or hyperchaotic behavior generated by the corresponding configurations has been verified in the relevant literature, thereby ensuring the fairness and consistency of the comparative experiments. Two sets of experiments are designed, with the signal-to-noise ratio (SNR) and the spreading factor (M) serving as the test variables, respectively. Furthermore, in the performance evaluation experiments for bit error rate (BER), 15 independent trials were conducted for each SNR or M , and the average of the obtained BER values was taken as the final result.

When the spreading factor is set to $M = 250$, Fig. 14(a) illustrates the BER performance of the HE-DCSK system utilizing different chaotic maps under different SNR. The experimental results indicate that as the SNR increases, the HE-DCSK system constructed with DMCHNN consistently achieves the lowest BER, outperforming all other comparative maps. Furthermore, under the condition of SNR $= 20$, Fig. 14(b) compares the impact of varying M on the system BER. It can be observed that the HE-DCSK system employing DMCHNN as the chaos generator exhibits the best anti-noise performance across the entire tested range.

The above experiments verify that the DMCHNN possesses superior BER performance in the HE-DCSK system, demonstrating its enhanced anti-noise and reliability in secure communication applications.

5. Conclusion

This paper proposes a discrete cosine memristor model and embeds it into a three-neuron Hopfield neural network with a cyclic structure, constructing a DMCHNN. Analysis of the system's fixed points and stability reveals the existence of countless unstable fixed points. Under the modulation of the coupling strength k and the weight w_{31} , the chaotic range of the system can achieve effective amplitude control, and its dynamical complexity is also enhanced. In particular, by adjusting the coupling strength k , the system can generate multi-cavity-like hyperchaotic attractors with varying numbers of cavities. Further investigation indicates that changing the initial conditions of the memristor can induce homogeneous/heterogeneous initial offset-boosting behavior, leading to the coexistence of innumerable homogeneous and heterogeneous attractors. Through a comparison with existing chaotic maps using multiple entropy metrics, the chaotic sequences generated by the system demonstrate superior entropy performance, indicating a higher degree of randomness. The implementation results based on the FPGA hardware platform verify the feasibility of DMCHNN, with experimental observations showing a high degree of consistency with numerical simu-

lations. In the application of the HE-DCSK communication system, the system constructed with DMCHNN exhibits superior anti-noise performance compared to other chaotic maps. Future research will focus on constructing high-dimensional discrete memristive neural networks with multiple LEs and further exploring their practical applications in information encryption systems.

CRedit authorship contribution statement

Gang Yang: Writing – original draft, Supervision, Methodology, Investigation, Data curation; **Chunhua Wang:** Visualization, Validation, Supervision, Funding acquisition, Conceptualization; **Quanli Deng:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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